

ECE250 Preliminary Project: Data

In this course you will read data for a time series and perform common statistical analyses on that time series. This is a preliminary project to ensure you can read your dataset.

Understanding the Data

The data is in comma-separated-values (CSV) format. You can open it in a spreadsheet program (like Excel or Google Sheets) to see what it looks like. In this course, all data is for countries with a name and Code, then series with a Name and Code, and finally for all years from 1960 to 2023, inclusive. Here is a sample:

Country Name	Country Code	Series Name	Series Code	1960	1961
Canada	CAN	Access to...	EG.ELC.ACC.ZS	..	25.3

Note: we have truncated the Series Name string. Here we see data for Canada, whose country code is CAN. The series name for this data is actually “Access to electricity (% of population)”, although we have truncated it in this document for brevity. The series code is EG.ELC.ACC.ZS, and for the year 1960 there is missing data (indicated by the string ‘..’) whereas for 1961 the value of the data is 25.3.

All data for all projects will follow this format: All rows consist of four strings followed either by some number of floating-point numbers or the exact string ‘..’ used as a value to indicate missing data. The number of columns is set by the number of projects in the first row. All of this data is separated by commas. **You may assume that no commas appear anywhere else in the dataset except to delimit the fields of the data.** You should open a data file and look at it to understand the format. You will be working with these files for the entire course.

Preliminary Exercise

It is extremely important to be able to read this data correctly, since all of our projects depend on it. Reading CSV files in C++ can be done in several ways, but the simplest is probably to use the `std::getline()` function and then the `std::stringstream` class to parse the data. We are going to parse the data for a single file.

Write a C++ program that achieves the following functionality:

1. Includes both `iostream` (for `getline`) and `sstream` (for `stringstream`)
2. Opens the file “project0_test_data.csv”. That file consists of several lines, each containing data as described above.
3. implements the following algorithm:
 - a. Using `std::getline`, read an entire line of the file into a string
 - b. Create a new `stringstream` object using the string you just read in as the argument to its constructor
 - c. Use `std::getline` again to extract each component of that string, passing in the comma as a delimiter
 - d. Print each component of the original string to the screen, one component per line

- e. Repeat for all lines in the file **without counting the number of lines beforehand**

A good starting point for this is here: <https://www.tutorialspoint.com/parsing-a-comma-delimited-std-string-in-cplusplus>, but note that in that tutorial they use the <vector> class, which you are not allowed to do. Rather than reading into a vector, simply print as soon as you extract the component.

Do not move on until you have mastered this functionality! It is crucial to the entire project!